Public Comment Letter to CMS

6/13/2025

Centers for Medicare & Medicaid Services
Department of Health and Human Services
Attention: CMS-0042-NC
P.O. Box 8000
Baltimore, MD 21244-8000

RE: Public Comment on Request for Information; Health Technology Ecosystem
2025-09701 (90 FR 21034)
RIN: 0938-AV68

Submitter Information

Name: Dr. Akram Boutros, MD
Title: CEO, Co-Founder
Organization: Nexus Bedside, Inc.
Address: 7800 Congress Avenue, Boca Raton, FL 33487
Email: aboutros@nexus-rn.com
Phone: 216-304-8440

Executive Summary

This response addresses the Centers for Medicare & Medicaid Services (CMS) and Assistant Secretary for Technology Policy/Office of the National Coordinator for Health Information Technology (ASTP/ONC) Request for Information on Health Technology Ecosystem, specifically focusing on hybrid-virtual nursing implementation in both Inpatient Prospective Payment System (IPPS) hospitals and Critical Access Hospitals (CAHs). Hybrid-virtual inpatient nursing, also known as a Virtually Integrated Care team, represents a transformative digital health solution that addresses workforce challenges while improving patient care quality, safety outcomes, and operational efficiency.

- For both Inpatient Prospective Payment System acute care hospitals and Critical Access Hospitals, nursing labor costs are the largest expense per discharge for the facility and therefore for CMS.

- Cost pressure can force hospitals to choose between clinically optimal care and ensuring financial and budget stability.
- Nurse staffing levels (e.g. the ratio of patients to nurses), skill mix, and the quality of nursing care are associated with adverse events, hospital acquired infections, falls, readmissions, and mortality–all of which are costly to CMS.
- Hybrid-virtual inpatient nursing services that leverage artificial intelligence technologies can reduce the cost of nursing care on a given unit, enable safer and more effective care services, and ultimately save money for hospitals and CMS.
- Hybrid-virtual inpatient nursing services are an emerging technology and process innovation that allow hospitals to redesign nursing workflows and staffing models to modernize hospital nursing care.
- Hybrid-virtual inpatient nursing allows nurses to rotate between virtual patient care and patient management shifts and in-person bedside shifts to improve job satisfaction and modernize workflows within care facilities.
- CMS should seek to incentivize hospitals to adopt hybrid-virtual nursing models of care as a policy intervention to address nursing workforce issues, patient care quality, and as a mechanism to reduce hospital expenditures.
- CMS should actively seek to fund deployment of hybrid-virtual inpatient nursing in critical access hospitals.

Background

Recent evidence from a systematic review of the evidence for the relationship between nurse staffing and certain patient outcomes, suggests important associations[1]. Most studies demonstrated that higher registered nurse (RN) staffing levels were associated with reduced patient mortality. Twelve analyses from seven studies, including three at low risk of bias, found significant associations between higher RN staffing and reduced in-hospital mortality in adult general populations. In adult intensive care and hospital settings, higher RN staffing was associated with statistically significant reductions in infections. Higher RN staffing was linked to several positive outcomes including reduced adverse events (death, cardiac arrest, unplanned ICU admission), shorter hospital stays (0.23 days reduction per additional RN hour per patient day), and lower readmission risk.

The inpatient nursing labor market is rife with key issues including high rates of clinician burn-out, moral injury related to understaffing, and competition from other sectors that are increasingly hiring inpatient bedside nurses away. NASEM predicts nursing shortages will worsen due to population

[1] Dall'Ora C, Saville C, Rubbo B, Turner L, Jones J, Griffiths P. Nurse staffing levels and patient outcomes: A systematic review of longitudinal studies. International Journal of Nursing Studies [Internet]. 2022 Jun 16;134:104311. Available from: https://doi.org/10.1016/j.ijnurstu.2022.104311

growth and aging demographics, particularly affecting rural areas that already lack sufficient nurses[2]. The report also notes that the average age of registered nurses (RNs) has dropped to just under 44 years due to approximately 1.2 million baby boomer RNs retiring while younger nurses enter the profession. Hospital-based RNs are notably younger at 42.3 years compared to those in non-hospital settings at 47.0 years. Data shows younger nurses (under 35) are leaving the profession at high rates in 2021, which threatens long-term workforce sustainability.

The rural nursing workforce has contracted significantly, with the proportion of RNs practicing in rural communities falling from 17 percent in 2005 to 14.4 percent in 2018. This decline affected both hospital and non-hospital rural settings, with rural hospital RN representation dropping from 16.4 percent to 13.4 percent, while rural non-hospital RN employment decreased from 18 percent to 16 percent during the same period[3]. The trend was particularly pronounced among younger nursing professionals, with RNs under 40 years old showing a steeper decline in rural practice (from 18.1 percent to 13.7 percent) compared to their older colleagues over 40 (from 16.4 percent to 14.9 percent). This demographic shift raises concerns about healthcare accessibility in rural communities and those with a supply demand imbalance for inpatient nursing labor.

The implications extend beyond current staffing challenges. With over 300,000 RNs employed in critical access hospitals (CAHs) and ongoing threats of rural hospital closures, the nursing shortage in these areas may intensify throughout the decade. This trend could undermine policy efforts designed to expand healthcare access for rural populations, potentially creating a cascade effect that further reduces both nursing and physician availability in underserved rural regions.

The job of the inpatient nurse itself has evolved with medical technology and with a growing body of evidence-based practice, however, the standard job itself, job environment, and workflows of the inpatient nurse have not been redesigned for decades. The addition of electronic medical records in the early 2000s significantly changed inpatient nursing care, yet nurses still primarily work in-person on 12 hour shifts as they have done for decades. They are also asked to perform more complex tasks, procedures, and manage a more complex inpatient patient population. Interruptions, for example, during patient care workflows are a critical issue for inpatient clinicians where one study suggests that nurses interrupted 12 times made twice as many clinical errors as nurses interrupted three times[4]. It is time to redesign the care delivery model of inpatient nursing care not only through minimally ensuring adequate staffing, but by rethinking the use of technology and workflows performed by the nursing care team.

---

[2] The Future of Nursing 2020-2030: Charting a Path to Achieve Health Equity. National Academies of Sciences, Engineering, and Medicine.;Washington DC:; National Academies Press; 2021. ISBN:;9780309685061

[3] The Future of Nursing 2020-2030: Charting a Path to Achieve Health Equity. National Academies of Sciences, Engineering, and Medicine. Washington DC:; National Academies Press; 2021. ISBN:;9780309685061.

[4] Santomauro C, Powell M, Davis C, Liu D, Aitken LM, Sanderson P. Interruptions to intensive care nurses and clinical errors and procedural failures: a controlled study of causal connection. *J Patient Saf*. 2021;17(8):e1433-e1440. doi:10.1097/PTS.0000000000000528

Technology, and a modern understanding of professional inpatient nursing practice, allow for the redesign of inpatient nursing care workflows. Artificial intelligence and an advanced hardware-software combination system operated by virtual inpatient nursing companies (e.g., Nexus Bedside) allows for the redesign of inpatient nursing workflows to reduce operating costs, improve quality patient outcomes, and improve workplace satisfaction among nursing staff. The model also allows for hospitals in low-nurse labor supply areas to access additional labor resources from other geographic areas in the United States.

A hybrid-virtual inpatient nursing care model implemented at an IPPS hospital or Critical Access Hospital pairs remote virtual nurses with bedside nursing teams to address nursing shortages and support inexperienced staff. Virtual nurses, working from remote locations, handle documentation-intensive tasks like admissions, discharges, medication reconciliation, and continuous patient monitoring using AI alerts and early warning systems, while bedside nurses focus on hands-on patient care. This team-based approach, guided by high reliability organization principles, has demonstrated significant improvements in patient safety through early detection of deterioration and reduced readmissions, standardized quality processes, optimized workflow efficiency, and maintained workforce stability during the COVID-19 pandemic. The program allows all team members to work at the top of their licensure scope while providing mentorship to novice nurses and creating a sustainable model that enhances both clinical outcomes and operational effectiveness.

We estimate that nursing represents 30% of hospital operating costs so CMS pays between $270 and $450 billion dollars per year on hospital nursing care alone per 2023 spending data. CMS payment policies and hospital regulations shape inpatient nursing care delivery models and modernization using technology is necessary. By supporting hospital adoption of the technology infrastructure for virtual nursing and enabling comprehensive and efficient outsourced management models under existing payment systems, the Inpatient Prospective Payment System (IPPS) and in payment to Critical Access Hospitals, CMS can improve the efficiency of inpatient care delivery and support rural hospital facilities in maintaining high standards of care while reducing operating expenditures.

Relevant CMS RFI Questions Addressed

This response directly addresses the following questions from the CMS RFI:

Provider Section:

- PR-1: Encouraging provider adoption of digital health products, including rural areas
- PR-2: Obstacles to development and deployment of innovative applications for physician workflows
- PR-7: Strategies to support providers in making healthcare data available for interoperability

Value-Based Care Section:

- VB-1: Incentives for APMs to leverage digital health management products
- VB-3: Essential health IT capabilities for value-based care arrangements

- VB-8: HHS policies to optimize digital health products in APMs

Technology Vendors Section:

- TD-1: Steps to stimulate developer interest in digital health products for Medicare beneficiaries

1. Definition and Scope of Hybrid-Virtual Nursing

Digital Health Product Definition (PC-1a, PR-1):

Hybrid-virtual inpatient nursing is a digital health technology-enabled solution that integrates remote nursing capabilities with bedside care delivery. This model employs registered nurses working from centralized virtual command centers who collaborate with on-site nursing teams to provide comprehensive patient care through audio-visual technology platforms embedded in patient rooms. The most advanced technologies incorporate artificial intelligence technologies integrated with the electronic health record and remote vital monitoring systems to automate repetitive nursing workflows, to catch potential clinical quality issues, and to enhance collaboration between the virtual and in-person nursing team as well as advanced practice nurses and physicians.

This model of care may be operated by the hospital via vendor relationships with a technology company or it may be entirely outsourced to a company that manages hiring, training, staffing, and technology implementation.

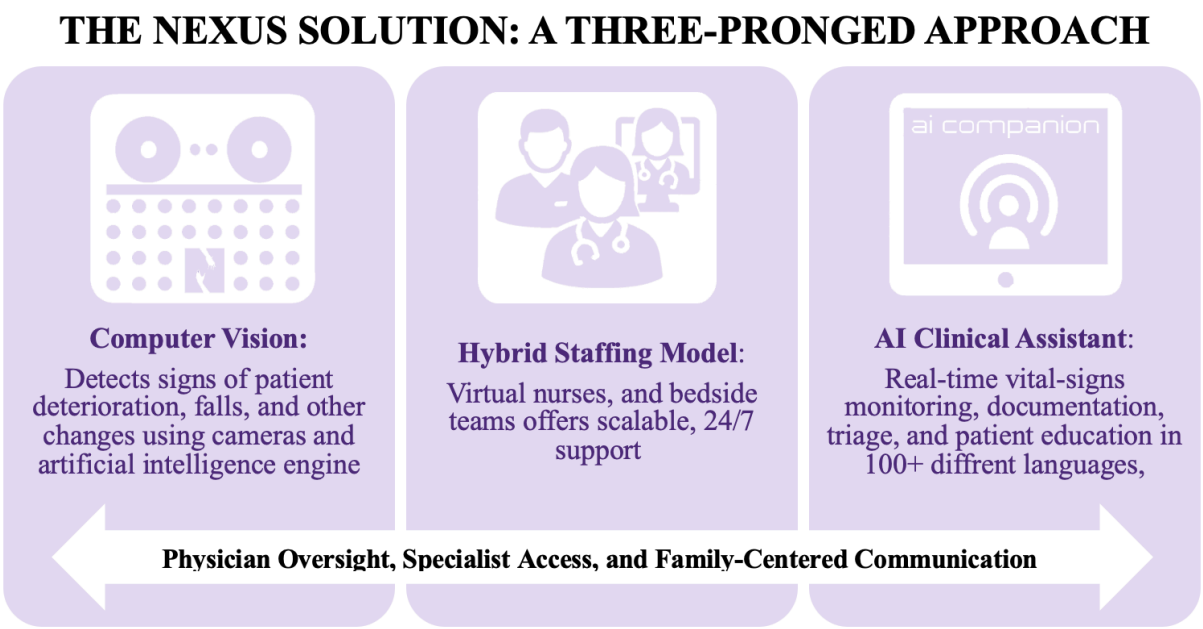
Core Technology Components:

- High-definition cameras and monitors in patient rooms
- Two-way audio communications technology between the patient, in-person care team, and off-site virtual nurse
- Secure, HIPAA-compliant communication platforms
- Integration with Electronic Health Records (EHR) systems
- Real-time access to patient data and monitoring systems
- Standardized workflow protocols for virtual and bedside nurse collaboration
- Artificial intelligence systems with both patient and nursing interfaces

Key Functional Modules

Virtual Nursing Station	Licensed nurses provide remote admission, education, discharge planning, and documentation support
Computer Vision AI	Passive continuous monitoring of patient movement, fall risk behaviors, and unsafe conditions (SaMD pathway as applicable)
Clinical Companion AI	EMR-integrated documentation assistance, progress note generation, early warning notifications
Patient & Family Portal	Real-time care plan visibility, educational content, test results, and secure two-way communication
Team Collaboration Tools	Secure HIPAA-compliant messaging, task management, care handoff coordination
Staffing Optimization Engine	Dynamic workload balancing and acuity-adjusted assignments
Analytics Dashboard	Continuous performance monitoring, QI reporting, and CMS-aligned outcome metrics

Figure 1. The Nexus Bedside Hybrid-Virtual Inpatient Nursing Approach



A Case Study of *Mercy Hospital's* Virtual Nursing Model

Mercy Virtual Nursing addresses the critical nursing shortage crisis through an innovative telehealth care delivery model. With nursing demand expected to increase by 21% by 2025, according to HHS, while 1 million nurses retire, Mercy created a 125,000 square foot Virtual Care Center that operates "like a

hospital without beds.[5]" This facility employs over 330 virtual care professionals who use high-definition cameras, real-time vital sign monitoring, and integrated electronic medical records to provide remote nursing care across 5 states and more than 30 intensive care units. The virtual nurses collaborate with bedside teams to provide 24/7 coverage, data synthesis to identify patient deterioration trends, mentorship for novice nurses, and documentation support. This model is very similar to that operated by Nexus Bedside, but Nexus leverages artificial intelligence to further enhance workflows, reduce missed care opportunities, and to ensure extra monitoring of patient safety.

The model has demonstrated **significant clinical improvements, including reducing septic shock progression from 34% to 19%** through early intervention based on automated data trending[6]. Virtual nurses maintain the critical nurse-patient relationship through video conferencing while providing expert assessment and care coordination. Additional technologies like mobile health devices enable continuous biometric monitoring of patients even after discharge, preventing readmissions and supporting care in patients' homes. The telestroke program brings specialist neurological care to community emergency rooms that lack onsite neurologists.

Rather than replacing traditional nursing, this telehealth model integrates technology to enhance and extend nursing care, particularly benefiting rural and underserved areas. It allows experienced nurses to continue practicing despite physical limitations, provides cost-effective care delivery, and maintains the compassion and therapeutic relationships essential to nursing while improving patient outcomes and satisfaction.

2. Benefits for IPPS Hospitals and Critical Access Hospitals

2.1 Addressing Staffing Challenges (PR-1a, VB-1)

Current Market Obstacles Addressed

Virtual nursing programs offer significant benefits for both IPPS hospitals and Critical Access Hospitals by directly addressing the persistent staffing challenges that threaten quality patient care across healthcare settings. These solutions tackle the widespread nursing workforce shortages and labor market challenges that affect facilities of all sizes, with rural facilities facing particularly acute pressures that are expected to intensify over the coming decade.

The recruitment and retention difficulties that plague rural healthcare facilities create ongoing operational instability and limit access to experienced nursing care. These challenges are compounded by broader demographic trends, including the retirement of experienced nurses, which has reduced the average years of clinical experience in inpatient facilities and potentially compromised care quality. Virtual nursing addresses the physical demands that traditionally limit career longevity for experienced nurses by creating

---

[5] Klingensmith, Liz, and Linda Knodel. 2016. “Mercy Virtual Nursing: An Innovative Care Delivery Model.” *Nurse Leader* 14(4):275–79. doi: 10.1016/j.mnl.2016.05.011.

[6] Ibid. Klingensmith 2016

opportunities for seasoned professionals to continue contributing their expertise in less physically demanding roles.

For Critical Access Hospitals specifically, virtual nursing solutions resolve the persistent challenge of providing limited 24/7 specialist nurse coverage. These rural facilities can now access specialized nursing expertise around the clock without the prohibitive costs of maintaining full-time specialist staff on-site, ensuring that patients receive appropriate care regardless of the time of day or the facility's geographic isolation.

Solution

Hybrid-virtual inpatient nursing with artificial intelligence augmentation helps solve these key policy obstacles. The solution leads to the following affects:

- Redesigns inpatient nursing workflows to improve quality, team coordination, patient experience, and discharge processes over and above legacy nursing models of care
- Enables experienced nurses to continue practicing without physical bedside demands
- Provides rural CAHs access to specialized nursing expertise from urban centers
- Supports 24/7 nursing coverage through centralized virtual nursing centers
- Allows one virtual nurse to support multiple patient rooms simultaneously
- Allows hospitals in rural areas or those with challenges obtaining nursing labor to hire and leverage nursing workforces in areas with a surplus

2.2 Quality and Safety Improvements (VB-3, PR-7)

Virtual nursing programs deliver substantial quality and safety improvements through essential health IT capabilities that enhance patient care across multiple dimensions[7]. Real-time patient monitoring creates continuous surveillance capabilities that effectively complement bedside nursing, ensuring that potential issues are identified and addressed promptly. Virtual nurses establish standardized care delivery protocols that ensure consistent admission processes, discharge procedures, and medication reconciliation across all patients, reducing variability that can lead to errors or oversights.

The quality assurance benefits extend to behind-the-scenes monitoring of critical interventions and documentation compliance, creating additional safety nets that catch potential problems before they impact patient outcomes. Virtual nursing also enables dedicated patient education time, allowing for comprehensive discharge education and thorough care plan discussions that might otherwise be rushed due to bedside nurses' competing priorities.

Documented Outcomes from Hybrid-Virtual Inpatient Nursing:

---

[7] Khairat, Saif, Jennifer Morelli, Qiyao Qin, Xiaoyu Wu, Randy Fakhreddin, Barbara S. Edson, and Mauri Williams. 2025. "Virtual Nursing's Effect on ED Efficiency and Quality of Care." *The American Journal of Emergency Medicine* 91:59–66. doi: 10.1016/j.ajem.2025.02.024.

Patient Safety Excellence: The most important benefit found in one program, similar to Nexus Bedside, was enhanced patient safety through early detection of patient deterioration[8]. Virtual nurses proactively activated rapid response interventions using Modified Early Warning Scores (MEWS), vital signs monitoring, lab values, and AI alerts. **This approach decreased Code Blue events outside the ICU, avoided unnecessary ICU transfers, and lowered overall mortality rates**.

Reduced Readmissions: Virtual nurses' involvement in discharge processes and patient education significantly reduced 30-day readmissions, with substantial cost avoidance benefits[9]. Under the CMS Hospital Readmission Reduction Program, reductions in readmissions are incentivized and provide a pathway to direct cost savings for CMS.

Care Standardization: Virtual nurses standardized admission, discharge, and transfer processes while maintaining high-quality outcomes[10]. They led medication reconciliation, progressive mobility assessments, and device utilization surveillance, reducing hospital-acquired infections through improved sepsis bundle compliance. These outcomes are directly inline with CMS quality monitoring goals and the Hospital Value-based Purchasing Program.

Clinical Documentation: The program enhanced documentation compliance and provided continuous surveillance of critical interventions, ensuring secure pathways for optimal patient care[11].

Reductions in Patient Falls: One 2024 study suggests that virtual nursing models can reduce patient falls by over 50%[12].

Reductions in Registered Nurse Turnover Rates: A 2024 study suggests that the integration of a virtual nursing work option for Registered Nurses can reduce turnover by 47 percent[13].

2.3 Operational Efficiency (TD-1, VB-8)

Workflow Optimization

Virtual nurses streamline healthcare delivery by taking on time-consuming administrative responsibilities such as admission documentation and discharge planning, which allows bedside nurses to dedicate their attention to direct patient care and clinical interventions. This division of labor creates faster response times to patient calls and questions through dedicated virtual access points. The efficiency gains extend to improved bed turnover rates as discharge processes become more streamlined, and healthcare facilities

---

[8] Sagastume, R., Peterson, J., (May 31, 2023) "The Virtual Nurse Program in a Community Hospital Setting" *OJIN: The Online Journal of Issues in Nursing* Vol. 28, No. 2, Manuscript 2.

[9] Ibid. Sagastume 2023
[10] Ibid. Sagastume 2023
[11] Ibid. Sagastume 2023
[12] Huun, Kathleen, and Rachel Spalding. 2024. “Virtually Integrated Nursing Care: A Case Study in Diffusion of Innovations.” *Management in Healthcare.* 9(1):79. doi: 10.69554/qjtl6989.

[13] Ibid. Huun 2024.

can operate with reduced staffing levels for a given patient census due to this hybrid approach that maximizes both virtual and in-person nursing resources.

The artificial intelligence system employed by Nexus Bedside further automates manual data entry, patient interaction summaries, and supports nursing workflows and care coordination among virtual nurses, the bedside care team, the patient, prescribing providers & specialists, and the patient's family. This system can allow nurses and clinicians to focus first on patient care rather than tedious and repetitive clinical workflows and tasks.

Cost Benefits

Healthcare organizations implementing virtual nursing models experience significant cost reductions through decreased dependence on expensive traveling nurses and overtime payments. This approach also reduces reliance on temporary staffing solutions, creating more stable workforce planning. The impact on nurse retention has been particularly notable, with a reported a 47% improvement in turnover rates[14]. These retention gains translate to lower recruitment costs as enhanced job satisfaction creates a more stable nursing workforce, reducing the ongoing expenses associated with constantly hiring and training new staff.

3. Implementation Considerations for Different Hospital Types

3.1 IPPS Hospitals

Implementation considerations vary significantly across different hospital types, with each requiring tailored approaches to maximize the benefits of virtual nursing programs.

IPPS hospitals are particularly well-positioned to implement virtual nursing solutions due to several inherent advantages. Their larger patient volumes create sufficient scale to justify the substantial technology infrastructure investments required for these programs. These facilities typically have established IT support systems that facilitate smoother implementation processes, along with greater financial resources that enable comprehensive virtual nursing program development. Additionally, IPPS hospitals can leverage their size and resources to serve as virtual nursing hubs, extending support to smaller facilities in their networks.

The implementation strategy for IPPS hospitals should focus on a phased rollout approach, beginning with specific units such as medical-surgical and telemetry departments where virtual nursing can demonstrate clear value. Integration with existing quality reporting requirements ensures that virtual nursing initiatives align with regulatory compliance and performance measurement systems. These larger hospitals can leverage economies of scale for technology procurement, reducing per-unit costs while ensuring robust system capabilities. Developing centralized virtual nursing centers allows IPPS hospitals to maximize their investment by supporting multiple facilities simultaneously, creating operational efficiencies and expanding their reach across broader geographic areas.

---

[14] Ibid. Huun 2024.

Cost Savings Opportunities for CMS

CMS yields cost savings from direct efficiency gains and downstream cost avoidance from higher quality care when hybrid-virtual inpatient nursing models are implemented augmented by artificial intelligence. First, a lower cost structure for this care delivery model will slow the growth of expenditures in inpatient care and show up first in the Medicare Cost Reports thus reducing the rate of inflation or generating cost savings in the IPPS Hospital Market Basket.

Second, better quality of care and lower rates of adverse events can reduce future cost to CMS both under the IPPS and under Medicare Part B after discharge.

Nexus Bedside targets the following key performance indicators in their program implementations:

Performance Metrics & CMS Outcome Alignment

CMS Priority Domain	Nexus Metric Target
Nursing Workload	30-40% reduction in bedside non-direct care task time
Patient Safety	40-60% reduction in falls, pressure ulcers, and preventable adverse events
Length of Stay	5-15% reduction through throughput optimization
Readmissions	10-20% reduction via improved discharge coordination
Care Experience	≥ 15% improvement in HCAHPS patient satisfaction
Nurse Retention	≥ 20% improvement in nurse engagement and turnover reduction
Financial Sustainability	15-30% labor cost stabilization for CAHs and rural providers

3.2 Critical Access Hospitals

Critical Access Hospitals face distinct challenges that make virtual nursing particularly valuable for their operations and patient care delivery. These facilities gain access to specialized nursing expertise that would otherwise be unavailable in their remote locations, while virtual nursing provides essential 24/7 nursing support despite their inherently limited on-site staffing capabilities. Virtual nursing solutions help CAHs maintain compliance with Conditions of Participation that require emergency care availability, ensuring they can meet regulatory requirements while managing resource constraints. Perhaps most importantly, these enhanced capabilities allow critical access hospitals to retain patients locally rather than transferring them to distant facilities, improving patient satisfaction and reducing the burden on patients and families.

The implementation landscape for CAHs presents both opportunities and challenges that require careful consideration. The cost-based reimbursement structure that provides 101% of reasonable costs creates a favorable environment for technology investments, as these expenses can be recovered through Medicare reimbursement. However, the upfront investment in a problem may require federal support. CAHs also have unique opportunities to share virtual nursing services across multiple facilities, creating cost efficiencies and ensuring sustainable access to specialized nursing expertise across rural healthcare networks.

4. Interoperability and Data Exchange Requirements (PR-5, PR-7)

4.1 Technical Standards Implementation

Successful virtual nursing implementation requires robust interoperability and data exchange capabilities that ensure seamless integration with existing healthcare technology infrastructure. The foundation of these technical requirements centers on comprehensive FHIR API implementation that enables virtual nurses to function effectively within the broader care team ecosystem.

The Patient Access API serves as a critical component, providing virtual nurses with comprehensive access to patient records necessary for informed clinical decision-making. The Standardized API for Patient and Population Services supports broader population health monitoring capabilities, allowing virtual nursing programs to contribute to quality improvement initiatives and outcome tracking. Provider Directory APIs facilitate essential communication pathways between virtual nurses and other healthcare providers, ensuring coordinated care delivery. SMART on FHIR integration enables virtual nursing applications to seamlessly integrate with existing EHR systems, maintaining workflow continuity and reducing technological barriers to adoption.

Data integration requirements extend beyond basic record access to encompass real-time monitoring capabilities essential for effective virtual nursing practice. Virtual nurses need immediate access to vital signs and monitoring data to provide timely interventions and support. Integration with nurse call systems and patient communication platforms ensures that virtual nurses can respond promptly to patient needs and maintain effective communication channels. Seamless documentation workflows across virtual and bedside nursing teams prevent information gaps and support comprehensive patient care coordination, while quality metrics tracking and reporting capabilities enable continuous program evaluation and improvement.

4.2 Privacy and Security Considerations

HIPAA Compliance:

Privacy and security considerations form the cornerstone of virtual nursing program compliance and patient trust. HIPAA compliance requires secure video conferencing platforms with end-to-end encryption to protect patient communications and maintain confidentiality. Access controls must limit virtual nurse

access to only their assigned patients, preventing unauthorized information exposure. Comprehensive audit trails for all virtual patient interactions ensure accountability and support compliance monitoring, while established patient consent processes for virtual nursing services maintain transparency and respect patient autonomy in care delivery decisions.

5. Specific Policy Recommendations (PC-5, VB-8, TD-1)

5.1 CMS Actions to Encourage Adoption

Reimbursement Considerations

- Clarify that hybrid-virtual inpatient nursing services are reimbursable under existing inpatient payment methodologies and included in Medicare cost reports for service lines that leverage these services and technologies
- Clarify that contracted, offsite, and generally supervised hybrid-virtual inpatient nursing services are reimbursable under the Critical Access Hospital cost-based reimbursement model
- Provide an incentive and upfront payment for both IPPS and Critical Access Hospitals to adopt hybrid-virtual inpatient nursing when hard costs savings are demonstrated from implementation
- Consider quality bonuses for facilities demonstrating improved outcomes through virtual nursing, perhaps through the Hospital Value-based Purchasing Program
- Provide added reimbursement at 110% of costs for critical access hospitals that adopt AI-nursing assistance technology as a part of a hybrid-virtual inpatient nursing model

CMMI Models

- Within IPPS hospitals, CMMI should provide upfront funds to hospitals for implementation of these programs when hard cost savings are demonstrated and able to be recouped by CMS under the model within three years. Improvements in quality of care and patient outcomes should provide a reduction in the cost-savings payback amount. We intend to submit a more detailed proposal on this model to CMMI.
- Within Critical Access Hospitals, CMMI should create a $100,000,000 program fund to be awarded to outsourced, vendor-managed hybrid-virtual inpatient nursing models to allow for rapid implementation of these models. We intend to submit further details on this program to CMMI.

Adoption Under Skilled Nursing Facility Prospective Payment System

- The minimum staffing standards set under CMS final rule CMS 3442-F for long-term care facilities indicates CMS goals of improving quality and safety in skilled nursing facilities. A hybrid-virtual model of nursing care is easily translated to this care setting to improve staffing while managing costs for these facilities

Quality Reporting Integration

- Incorporate virtual nursing metrics into Hospital Inpatient Quality Reporting (IQR) Program

- Develop quality measures specific to virtual nursing effectiveness

5.2 Regulatory Framework Enhancements

Information Blocking Prevention:

- Ensure virtual nursing platforms have appropriate API access to EHR systems
- Specifically clarify and enforce information blocking rules in the context of data sharing with inpatient virtual nursing providers and the corresponding technology systems employed
- Support standardized data exchange protocols for virtual nursing

Certification Requirements:

- Develop ONC Health IT Certification Program criteria for virtual nursing platforms
- Ensure interoperability standards support virtual nursing workflows
- Include virtual nursing capabilities in Certified Electronic Health Record Technology (CEHRT) requirements

6. Rural Hospital Considerations (PR-1a, VB-10)

6.1 Critical Access Hospital Specific Benefits

Rural hospitals, particularly Critical Access Hospitals, face unique operational challenges that make virtual nursing solutions especially valuable for maintaining quality care, access to nursing labor, and financial sustainability. These facilities must navigate the complex requirements of providing comprehensive healthcare services in resource-constrained environments while serving geographically dispersed populations in areas with limited nursing labor markets.

Virtual nursing provides essential staffing flexibility that directly addresses the regulatory and practical challenges facing Critical Access Hospitals. These facilities must comply with CAH requirements for 24/7 emergency care availability, which can be difficult to maintain with limited nursing staff. Virtual nursing solutions enable appropriate RN coverage during periods when only Licensed Practical Nurses are physically present on-site, ensuring that patients receive the appropriate level of clinical oversight and intervention capabilities. Additionally, rural hospitals gain access to specialized nursing expertise in areas such as ICU care, cardiac monitoring, and oncology support that would otherwise be unavailable due to geographic isolation and limited resources. Virtual nurses that service these facilities may be located in a distant area, but can be licensed in the same state in which they are providing care to the patient.

The financial sustainability benefits of virtual nursing align particularly well with the unique reimbursement structure of Critical Access Hospitals. The cost-based reimbursement model that provides 101% of reasonable costs creates a favorable environment for virtual nursing technology investments, as these expenses can be recovered through Medicare reimbursement. Virtual nursing reduces the need for expensive contingent labor from travel nurses or *locum tenens* arrangements for advanced practitioners. Perhaps most importantly, improved patient retention capabilities reduce revenue loss from transfers to

distant facilities, allowing rural hospitals to maintain their patient populations and the associated reimbursement while providing higher-quality local care that better serves their communities.

Within regions, hub-and-spoke models can be developed to allow for collaborative staffing arrangements with larger facilities. Groups of critical access hospitals can also share services to reduce the cost of implementation and to ensure the financial sustainability of these programs.

In the age of artificial intelligence, these facilities also may lack the infrastructure to adopt the very technologies that may benefit them most. CMS, HRSA, and HHS should carefully examine and consider specific hybrid-virtual nursing programs for critical access hospitals via payment models driven by the Center for Medicare and Medicaid Innovation.

7. Value-Based Care Integration (VB-1, VB-3, VB-8)

7.1 Value-based Care General Alignment

This model of care is well-aligned with value-based care and the Quadruple Aim in that it is designed to produce hard cost savings for facilities and produce better patient outcomes while supporting the nursing workforce. CMS realizes cost savings when reduced inpatient nursing costs are realized on the Medicare Cost Report on form CMS-2552-10. This may reduce hospital cost inflation and result in savings via the Hospital Market Basket and therefore in the following year's Operating Base Payment. CMS also accrues cost savings from reduced adverse events, falls, readmissions, and hospital use of contingent labor.

8. Research and Development Priorities (PC-7, TD-1)

8.1 Evidence Generation Needs

The continued advancement of hybrid-virtual inpatient nursing requires focused research and development efforts that address both immediate implementation needs and future innovation opportunities. Evidence generation represents a critical priority for establishing the long-term viability and effectiveness of virtual nursing programs across diverse healthcare settings. CMS, HHS, and the National Institutes of Health should fund patient-centered outcomes research on quality outcomes from these programs as well as cost-savings and return on investment to CMS.

Clinical outcomes research must examine the long-term impact of virtual nursing on patient safety metrics to establish definitive causal evidence of improved care quality. Cost-effectiveness studies comparing virtual nursing models to traditional staffing approaches will provide essential data for healthcare administrators and policymakers making investment decisions. Comprehensive patient satisfaction and experience measurement will ensure that virtual nursing enhances rather than diminishes the patient care experience, while research into nurse retention and job satisfaction outcomes will demonstrate the workforce benefits that can sustain these programs over time.

Future innovation opportunities center on artificial intelligence integration that can dramatically enhance virtual nursing capabilities. Predictive analytics for patient monitoring will enable proactive interventions

before clinical deterioration occurs, while automated alert systems can ensure rapid response to changing patient conditions. Natural language processing for documentation support can reduce administrative burden and improve accuracy, and machine learning applications can continuously optimize care protocols based on outcome data. Nexus Bedside's application of artificial intelligence in this particular area is quite promising and is already implemented at facilities.

Expanded virtual care models represent the natural evolution of virtual nursing beyond traditional hospital boundaries. Integration with hospital-at-home programs can extend virtual nursing support into patients' residences, while virtual specialist consultations can provide expert care access regardless of geographic limitations. Remote monitoring capabilities for post-discharge care can reduce readmissions and support care continuity, and comprehensive telehealth platform integration will create seamless care experiences that span multiple settings and provider types.

Conclusion

Hybrid-virtual nursing represents a transformative digital health solution that directly addresses the workforce challenges facing both IPPS hospitals and Critical Access Hospitals while improving patient care quality and operational efficiency. The technology aligns with CMS priorities for advancing interoperability, supporting value-based care, and leveraging digital health products to enhance Medicare beneficiary outcomes.

We recommend that CMS consider virtual nursing as a priority area for policy development, quality measurement, and reimbursement clarification. The evidence demonstrates significant potential for improving patient care, supporting nursing workforce sustainability, and advancing the goals of a patient-centered digital health ecosystem.

Respectfully submitted,

Dr. Akram Boutros, MD
CEO, Co-Founder
12th June, 2025

Dr. Akram Boutros, MD

Akram Boutros, MD, FACHE, has over 30 years of executive management experience. A distinguished healthcare executive, Dr. Boutros previously served as the CEO of The MetroHealth System in Cleveland, Ohio, where he was recognized for transformative leadership and innovation. He earned his Doctor of Medicine degree from Downstate Health Sciences University and holds a Bachelor of Science in Biology and Chemistry from St. John's University. Throughout his career, he has received numerous accolades, including being named one of *Modern Healthcare's* Most Influential Clinical Leaders. Akram Boutros, M.D., FACHE, is a transformative leader and one of the top operating executives in healthcare. Throughout his career, Akram has built high-performing, mission-driven organizations that enable people to grow and excel. He is an expert in navigating strategic inflection points, driving organizational reinvention, and fostering cultures that amplify frontline impact. His guiding principle: leading with love creates enthusiastic, dynamic, and enduring workplaces.

As CEO of The MetroHealth System, Akram achieved extraordinary results: tripling the number of patients served, doubling revenue, and increasing operating income tenfold. Under his leadership, MetroHealth was recognized as more socially responsible than 99.5% of U.S. hospitals and became the nation's top-performing safety-net Accountable Care Organization.

Earlier in his career, Akram excelled in academic medicine, focusing on graduate education and operational efficiency through analytics. He went on to lead community and specialty hospitals, emphasizing physician integration, centers of excellence, and business diversification. By 2008, he pivoted to clinical and operational transformation, mastering the strategies required for full-risk reimbursement models. For many years Akram also focused on technology-enabled care delivery. He founded several innovative organizations, including Doc2Go, CLE Care, Select Assurance, Spry, Spry Senior, Lobesity, Lumina, Ovatient, and Nexus Bedside.

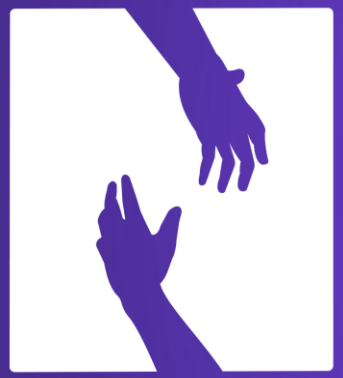
His contributions have earned him widespread recognition, including two honorary doctorates, Modern Healthcare's "Most Influential Clinical Leaders," humanitarian and innovator awards, and knighthood by the Vatican and the House of Bourbon. Akram holds an M.D. from SUNY Downstate and completed the Advanced Management Program at Harvard Business School.

Submission Information

Submission Method: Electronic via regulations.gov
Confirmation Number: [If submitting electronically]
Date Submitted: 15 June, 2025

Additional Documentation

- Attachment A: Nexus Bedside Presentation



Nexus Bedside

The future of inpatient nursing.

- **HYBRID INPATIENT CARE PLATFORM - CMS SUBMISSION**

WORKFORCE STABILIZATION

PATIENT SAFETY

CARE TRANSFORMATION

MISSION & VISION

MISSION

Empowering our healthcare partners with a tech-enabled care tram, elevating clinical outcomes & efficiency

+

VISION

To revolutionize inpatient care with an intelligent model that prioritizes nurses to achieve extraordinary performance

=

EXECUTIVE SUMMARY

Combines hybrid nursing, intelligent technology, real-time clinical coordination

Stabilizes workforce, improves outcomes, lowers costs

Fully aligned to CMS Innovation Center priorities

THE NATIONAL WORKFORCE CRISIS

1.0M inpatient nurse shortage by 2036

40% retiring within 5 years

HOSPITALS AT RISK OF CLOSURE

37-57% of new nurses leave within 2 years

Temporary labor costs surging

THE NEXUS SOLUTION: A THREE-PRONGED APPROACH

Computer Vision:

Detects signs of patient deterioration, falls, and other changes using cameras and artificial intelligence engine

Hybrid Staffing Model:

Virtual nurses, and bedside teams offers scalable, 24/7 support

AI Clinical Assistant:

Real-time vital-signs monitoring, documentation, triage, and patient education in 100+ diffrent languages,

Physician Oversight, Specialist Access, and Family-Centered Communication

CMS INNOVATION ALIGNMENT

Workforce Stabilization: Hybrid Nursing Model

Patient Safety: Ai-powered Surveillance

Health Equity: Rural & Underserved Hospitals

Financial Sustainability: Stabilized Labor Costs

Care Coordination: Virtual + Bedside Team Integration

HOW NEXUS WORKS (PLATFORM OVERVIEW)

- Virtual Nursing Station
- Vision AI Monitoring (SaMD-eligible)
- Clinical Companion AI
- Dynamic Staffing Engine
- Family Engagement Portal
- Central Control Room

REAL CLINICAL OUTCOMES

Nurse Overtime ↓ 30-40%
Falls/Adverse Events ↓ 40-60%
LOS Reduction ↓ 5-15%
Nurse Turnover ↓ 20%
Patient Satisfaction ↑ 15%
Documentation Timeliness ↑ 95%

HEALTH EQUITY & ACCESS IMPACT

Enables rural hospitals to maintain safe inpatient care

Supports safety-net hospitals under financial strain

Enhances Medicaid/Dual Eligible coordination

Enables remote, retired, or disabled nurses to return virtually

CMS FINANCIAL IMPACT

01

HAC Reductions: Lower penalties

02

Readmissions: Avoidance savings

03

LOS: Throughput gains

04

Agency Labor: Reduced dependency

05

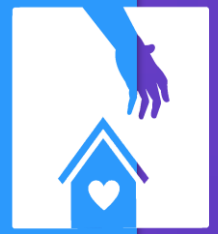
VBC Participation: Enhanced performance

FULL CONTINUUM CAPABILITY

Nexus Hospice Care: Hybrid inpatient & home hospice

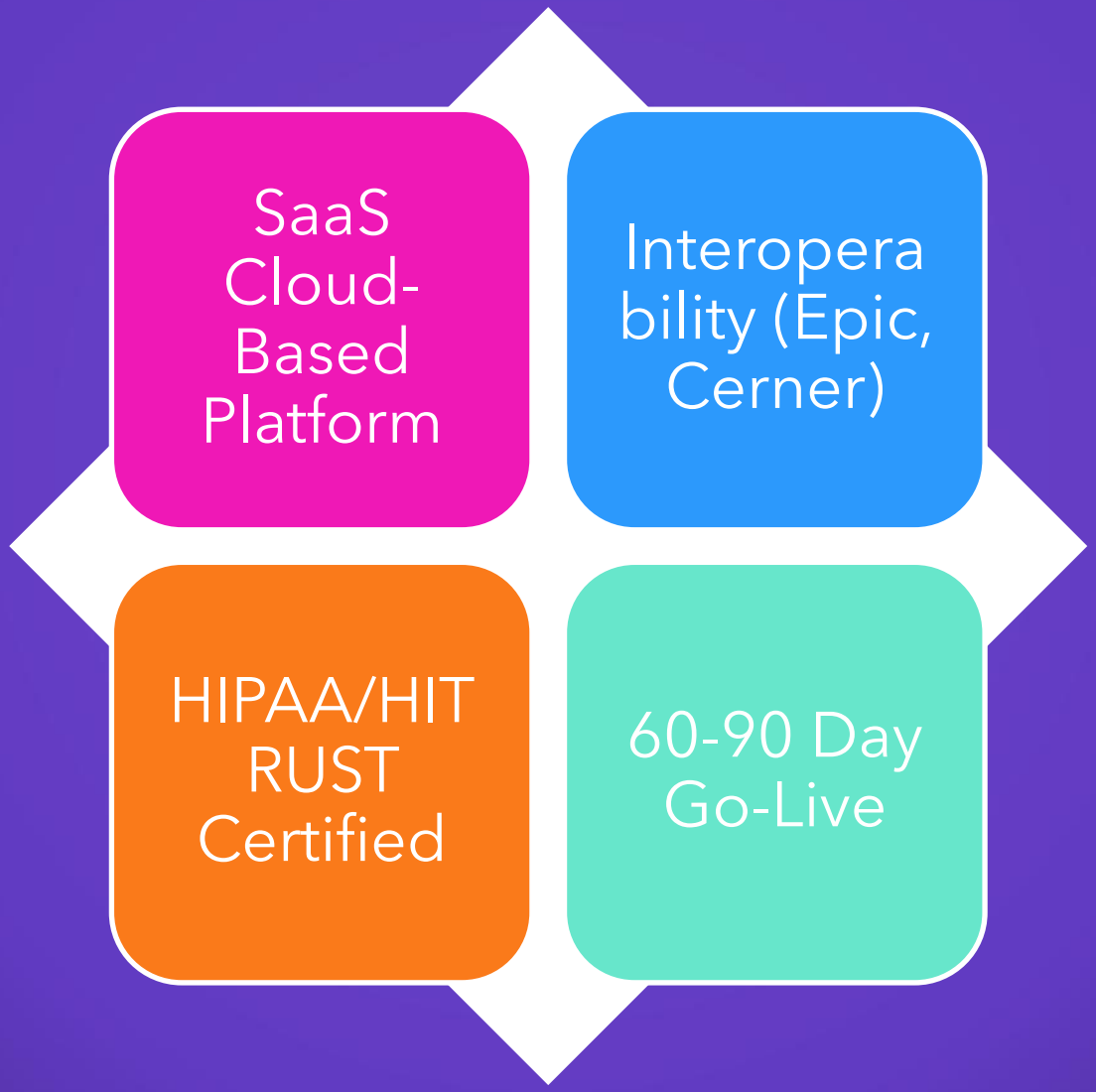
Nexus Hospice Care

Nexus At Home: Hospital-at-home for frail elderly, SNF diversion



Nexus at Home

DEPLOYMENT MODEL



Thank You!